



टेलीकम्युनिकेशन्स कंसलटेंट्स इंडिया लिमिटेड
(भारत सरकार का उद्यम)
Telecommunications Consultants India Limited
(A Government of India Enterprise)



**प्रभाग
DIVISION**

टी. सी. आई. एल. भवन, ग्रेटर कैलाश-I
नई दिल्ली-110048

TCIL Bhawan, Greater Kailash-I,
New Delhi-110048

CIN : U74999DL1978GO1008911

No. TCIL/SR/HYD/2024/T-Fiber/Exe (Vol-XII)

Dated: 20-05-2024

**TO WHOM IT MAY CONCERN
EXPERIENCE CERTIFICATE**

This Experience Certificate is issued to **M/s Quadgen Wireless Solutions Private Limited** on their specific request only vide letter QG/Corp/2024-25/003 dated 16/04/2024 as per details given below:

S. No.	Item	Description
1.	Name of Agency / Vendor	M/s QuadGen Wireless Solutions Private Limited.
2.	Name of Work / Project	Implementation of Package-C of Telangana Fiber Grid Project (T-FIBER) in the State of Telangana by Telangana Fiber Grid Corporation Limited
3.	Project Location	Telangana State – Package-C covering 10 Districts (Adilabad, Asifabad, Mancherial, Nirmal, Nizamabad, Kamareddy, Karimnagar, Rajanna Sircilla, Jagityal and Peddapalle).
4.	Client's Name	Telecommunications Consultants India Limited. (TCIL) for Telangana Fiber Grid Corporation Ltd. (T-Fiber), Telangana
5.	Type of work	Route Survey, Design, Planning of Network Architecture, Designing the solution, Finalization of BoQ, OFC Laying work, installation, testing and commissioning of active equipment, integration of different networks as well as integration with NOC, Operations and Maintenance of the end-to-end network infrastructure.
6.	TCIL's LOI / PO / WO No with Date	TCIL/15/1860/I/19-MM/75P dated 17.01.2020. TCIL/15/1860/1/19-MM/75P/AO-1 dated 13.10.2021 TCIL/15/1860/I/19-MM/75P/AO-2 dated 17.02.2022. TCIL/15/1860/I/19-MM/75P/AO-3 dated 21.04.2023. TCIL/15/1860/I/19-MM/75P/A-3A dated 04.07.2023.
7.	TCIL's LOI/ PO / WO Amount without GST	Rs 913,80,01,701/- (Rs Nine Hundred Thirteen Core Eighty Lakhs One Thousand Seven Hundred and One Only) for Phase –I including O&M.
8.	TCIL's LOI/ PO / WO Amount with GST	Rs 1078,28,42,007/- (Rs One Thousand Seventy Eight Crores Twenty Eight Lakhs Forty Two Thousand Seven only) for Phase –I including O&M.
9.	Date of commencement of work	22-01-2020
10.	Duration of work/ project	15 Months (Execution) and 7 Years (O&M)
11.	Actual Date of Completion	Ongoing – 2206 GPs Lit-up completed, 3.5 Districts are Live and under O&M. Scope of work: • GPs to connect = 3057 • Mandals to connect = 182

राष्ट्रीय : National : टेलीफोन : Telephone : 011-26202020 फैक्स : Fax : 011-26242266 वेबसाइट : Website : <https://www.tcil.net.in> ई-मेल : E-mail : tcil@tcil.net.in
अंतराष्ट्रीय : International : टेलीफोन : Telephone : +91-11-26202020 फैक्स : Fax : +91-11-26242266 वेबसाइट : Website : <https://www.tcil.net.in> ई-मेल : E-mail : tcil@tcil.net.in

हम पढ़ें सभी भाषाओं को, अपनी हिंदी का मान करें, सम्मान बढ़ाएं भारत का, आओ हिंदी में काम करें।



S. No.	Item	Description
		- Govt Institutes to connect = 7365 Work completed till date: - OFC laying = 7915 Km - Routers Installed = 659 - Metro Switches Installed = 643 - OLTs (16P) Installed = 264 - OLTs (8P) Installed = 340 - ONTs Installed = 3362
12.	AMC period	7 Years (O&M) Presently performing O&M for 2153 route Kilometers covering 818 GPs across 91 Mandals since 16-07-2022.
13.	Value of work done till date including LD / Penalty without GST	Rs 652,20,04,194/- (Rs. Six Hundred Fifty Two Crores Twenty Lakhs Four Thousand One Hundred Ninety Four only)
14.	Value of work done till date excluding LD / Penalty without GST	Rs 652,20,04,194/- (Rs. Six Hundred Fifty Two Crores Twenty Lakhs Four Thousand One Hundred Ninety Four only)

Note:

- 1) This Experience Certificate is not a declaration or confirmation of any final payment made by TCIL to the said vendor. TCIL reserves the right to withhold or deduct any amount pertaining to penalty / LD / taxes etc., which may be found applicable, from any payments to be made to the vendor from time-to-time in future or at the time of full & final settlement.
- 2) This Letter of Experience is being issued upon specific request from M/s QuadGen Wireless Solutions Private Limited for participating in the BSNL released amended BharatNet Tender only.


 Sanjay Kumar
 CGM (SCM)





Telecommunications Consultants India Ltd.
(A Govt. of India Enterprise)
Material Management Division
TCIL Bhawan, GK-I, New Delhi – 110 048
Telephone no.:011-26202020, Fax:011-26241865
Website: www.tcil.net.in
CIN No.: U74999DL1978GOI008911



TCIL/15/1860/I/19-MM/75P/AO-3A

04.07.2023

AMENDMENT-3A TO LETTER OF AWARD

M/s QUADGEN Wireless Solutions Private Limited (QUADGEN)

#607, World Trade Center, Brigade Gateway
No. 26/1 Dr. RajKumar Road, Mallesaram West,
Banagalore-560055, India

Kind Attention: Mr. Satish Benarji, COO

Dear Sir,

Sub: Implementation of Package-C of Telangana Fiber Grid Project (T-FIBER) in the State of Telangana by Telangana Fiber Grid Corporation Limited (T-FIBER LTD.)

- Ref:
- 1) RFP documents vide TSTS/CS/RFP-T-FIBER-MSI/2018 dated 20.03.2018 along with its corrigendum, addendums & errata.
 - 2) TCIL EOI No. TCIL/TG/044/2018/1/MSI-Telangana dated 17.05.2018 along with its corrigendum, addendums & errata.
 - 3) M/s QUADGEN's offer in a consortium bid with M/s CEIBA ECC Private Limited and M/s EMC Limited against TCIL's EOI for the subject tender, submitted vide email dated 17.09.2018.
 - 4) QUADRIPARTITE AGREEMENT dated 19.09.2019 signed among TCIL, QUADGEN, CEIBA ECC Pvt. Limited (CEIBA) and EMC Limited (EMC).
 - 5) M/s QUADGEN's revised price vide email dated 14.02.2019 based on L1 Price of M/s L&T.
 - 6) Signed MoM of meeting held on 05.12.2019 between T-FIBER LTD. & TCIL
 - 7) M/s QUADGEN's letter dated 29.12.2019 regarding undertaking for project phased approach & confirmation on validity of prices for 3 years.
 - 8) M/s QUADGEN's letter dated 31.12.2019 regarding price validity undertaking letter & acceptance of MoM for meeting held on 05.12.2019.
 - 9) TCIL Letter No. TCIL/76/001/IT-OFC/02/2019-20 dated 07.01.2020 for acceptance to T-Fiber LOA.
 - 10) TCIL LOA dated 17.01.2020 to M/s QUADGEN with its AO-1 dated 13.10.2021, AO-2 dated 17.02.2022 and AO-3 dated 21.04.2023.
 - 11) T-Fiber LOA dated 31.12.2019 with its Amendment-I dated 26.04.2021, Amendment-II dated 30.12.2021, Amendment-III dated 28.03.2023 and Amendment-IIIA dated 18.05.2023.
 - 12) TCIL Letter No. TCIL/15/PBG-Extn./MM/23-24/QWSPL/07 dated 12.06.2023.

Dear Sir,

1. LOA Amendment-3 was issued on back-to-back basis of LOA Amendment-III by T-Fibre, wherein new line items were added as per requirement of the project.



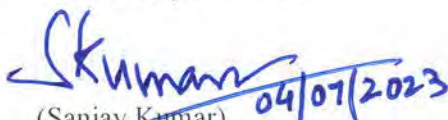
2. The client (T-Fibre), during processing of invoices, observed that Sl. No. in the LOA Amendment-II and III have some variations and has issued Amendment-IIIA after corrections in such Sl. Nos.
3. All other terms & conditions mentioned in AO-3 dated 21.04.2023 remain unchanged, including scope of work, deliverables, timelines, and financial provisions. This amendment solely focuses on amending the serial numbers without any financial implication to ensure clarity and accuracy in billing process.

4. **Contract Price:**

Subsequent to AO-3 dated 21.04.2023, the contract price to carry out the work as per scope of T-Fiber project's Package-C is **Rs. 913,80,01,701/- (Rupees Nine Hundred Thirteen Crore Eighty Lakh One Thousand Seven Hundred one only)**, all-inclusive except GST, as per enclosed Annexure-1.

5. M/s QUADGEN is yet to submit the revised Performance Security, as per AO-3 terms & conditions (10% of CAPEX Value, amounting to Rs. 70,00,20,424/-). It is requested that M/s QUADGEN submit the revised performance security of value **Rs. 70,00,20,424/-** valid up to **31.12.2023**, latest by 12.07.2023 as per TCIL letter reference cited at 12 above.

The copy of this Amendment-3A to LOA duly stamped and signed by the authorized person be returned within 2 days of issuance of this LOA as a token acceptance of this LOA, failing which it will be presumed that this LOA has been accepted in total.


(Sanjay Kumar) 04/07/2023
Chief General Manager (SCM)

Encl: Revised Annexure 1 - BOQ & Price

Copy To:

1. ED (IT&T-III), TCIL
2. CGM (F&A), TCIL
3. Master File
4. Office copy



Annexure 1 (BOQ and Price) -LOA Amendment - 3

S. No	Item Description	Unit	AO-3 Qty	Unit Cost (INR)	AO-3 cost(INR)
	SHQ to GP				
B.1	Physical Survey				
1	Physical Survey of route, capturing the latitude- longitude of optical fiber cable route , route & joint indicator, preparation of GIS survey reports , integration of GIS data with as build diagram of OFC route -From SHQ till GP	Km	11,048		
2	Physical Survey of route, GP & Testing for taking over for NOFN Phase – I @ 2.25 kms (on average) per PoP	No.	729		
3	DGPS Survey in Forest Area	Km	787		
B.2	Optical Fiber Cable				
1	Optical Fiber Cable 288 Core - Underground	Km	371		
2	Optical Fiber Cable 144 Core - Underground	Km	868		
3	Optical Fiber Cable 96 Core - Underground	Km	2,270		
4	Optical Fiber Cable 48 Core - Underground- Linear Portion	Km	7,761		
5	ADSS OFC 48 Core - For Ring Completion	Km	537		
6	Drop Cable 2 Core(Oval type) for Govt. Institution Connectivity	Km	7,905		
B.3	Underground OFC Accessories				
1	HDPE PLB Duct (40 mm / 33mm)	Km	4,706		
2	DWC Pipe (75/61 mm) @ 50Mtr per Km	Km	364		
3	GI Pipe 50mm - Class-B Medium as per ISI Mark@60 mtr per Km	Km	37		
4	Supply of Route / joint indicator	No	13,943		
5	Supply of Manhole & Joint chambers	No	12,766		
6	Supply of FDMS outdoor (Joint Closure) for 288 Core Fiber	No	473		
7	Supply of FDMS outdoor (Joint Closure) for 144 Core Fiber	No	843		
8	Supply of FDMS outdoor (Joint Closure) for 96 Core Fiber	No	1,930		
9	Supply of FDMS outdoor (Joint Closure) for 48 Core Fiber	No	4,583		
B.4	ADSS OFC Accessories				
1	Formed OFC Dead End & Termination Fittings @ 5 per Km (Wedge Type)	Set	2,457		
2	Formed OFC Dead End & Termination Fittings @ 5 per Km (Helix Type)	Set	613		
3	OFC Suspension Fittings @ 15 per Km (Wedge Type)	Set	7,370		
4	OFC Suspension Fittings @ 15 per Km (Helix Type)	set	1,825		
5	De-mountable pulley @ 1 per 10 Km	No	122		
6	Adjustable Cable Storage Bracket @ 3 per Km	Set	1,814		
7	Galvanized steel poles of tubular in design and swaged of the length 7 meters, with top cap welded and lightning spike and including MUFF @ 15 per Km.	No	3,564		
8	Galvanized steel poles of tubular in design and swaged of the length 8 meters, with top cap welded and lightning spike and including MUFF @ 2 per Km.	No	318		
9	Straight Joint Closure / Branch Joint Closure @ one per 2 Km	No	274		
10	Stay arrangement for 7M/8M steel poles	No	5,415		
B.5	Zone PoP				
1	Zone Routers (including SFP modules for connecting Zones to Mandals)	No	3		
2	Pizza Box type OLT (8 port) for Horizontal Connectivity	No	2		
3	Pizza Box type OLT (16 port) for Horizontal Connectivity	No			
4	24 Port Distribution Switch for connectivity to local end points	No	3		
5	FDMS Type-1 for Zone	No	3		
6	Air Conditioner - 2 Ton (Industrial Grade) (Zone)	No	6		
7	Rack - 42U floor mount with all accessories for Zones	No	9		
8	Online UPS (15 KVA)	No	3		
9	D.G Set (25 KVA)	No	3		
10	Servo Stabilizer(25 KVA)	No	3		
B.6	Mandal PoP				
1	Mandal Router (including SFP modules for connecting Mandals to GPs)	No	182		
1.1	10G BASE-LR SFP Module	No			
1.2	10G BASE-ER SFP Module	No	126		
1.3	10G BASE-ZR SFP 10G Module for SMF	No	38		
2	OLT for Horizontal Connectivity at Mandal PoP – Chassis Based	No	182		
3	OLT for Horizontal Connectivity at Mandal PoP – Pizza Box (8 Port)	No			



Annexure 1 (BOQ and Price) -LOA Amendment - 3

S. No	Item Description	Unit	AO-3 Qty	Unit Cost (INR)	AO-3 cost(INR)
4	OLT for Horizontal Connectivity at Mandal PoP – Pizza Box (16 Port)	No			
5	ONT	No	4,398		
6	FDMS (Indoor) at Mandal- 288F	No			
7	FDMS (Indoor) at Mandal- 144F	No	364		
8	FDMS (Indoor) at Mandal- 96F	No	59		
9	FDMS (Indoor) at Mandal- 48F	No	195		
10	Rack - 42U floor mount with all accessories for Mandals	No	182		
11	Online UPS (6 KVA) with 60 minutes power back up	No	182		
12	Servo Stabilizer (20 KVA)	No	182		
13	Air Conditioner - 2 Ton (Industrial Grade) (Mandal PoP)	No	182		
14	Point To Point/ Point To Multi Pont Links for wireless connectivity	No			
15	16 Port Distribution Switch for connectivity to local end points	No	182		
B.7 Gram Panchayat PoP					
1	Access Router (including SFP module for connecting OLT)	No	1,228		
1.1	10G BASE-LR SFP Module	No			
1.2	10G BASE-ER SFP Module	No	1,000		
1.3	10G BASE-ZR SFP 10G Module for SMF	No	28		
2	8 Port Distribution Switch for connectivity to local end points	No	1,228		
3	Metro Ethernet Switch (L3)	No			
4	Pizza Box Based OLT (8 Port)	No	874		
5	Pizza Box Based OLT (16 Port)	No	231		
6	Chassis Based OLT	No	123		
6.1	GPON card: 16xGPON interface module	No	100		
6.2	10G optical transceiver, SFP+, 1310nm, 10Gb/s, SM 10KM	No	350		
6.3	GE SX MM 850 nm pluggables	No	100		
7	Rack - 24U Floor mount with all accessories for GPs	No	3,786		
8	FDMS(Indoor) at GP- 24 F	No	491		
9	FDMS(Indoor) at GP- 48 F	No	3,057		
10	Online Hybrid UPS at GP PoP (1 KVA)	No	3,593		
11	HFTB (indoor)-Govt. Institutions	No	12,150		
12	ONT	No	7,365		
13	Point To Point/ Point To MultiPont Links for wireless connectivity	No			
14	Cat6 Eternet cable	Km	150		
15	RJ45 connectors	No	20,000		
B.8 EMS for NOC					
1	Element Management System (EMS) Solution including all Hardware, Software, Licenses, etc. (For IP-MPLS)	No	1		
2	Element Management System (EMS) Solution including all Hardware, Software, Licenses, etc. (For GPON)	No	1		
B.9 Services for UG OFC laying					
1	Duct Integration Testing (DIT) and rectification of the existing duct, where DIT fails and/or the duct is missing, including all the material and services required for rectification/replacement of missing duct	Km	5,568		
2	Excavation of trench and backfilling, reinstatement and compact of the trench for PLB pipe laying (by MSI) and duct laying	Km	4,706		
3	Pulling/laying / blowing of optical Fibre Cable (48 core fiber) inside laid PLB duct and at crossings(with protection provided) , splicing/jointing, commissioning, AT, ABD etc.	Km	7,751		
4	Pulling/laying / blowing of optical Fibre Cable (96 core fiber) inside laid PLB duct and at crossings(with protection provided) ,splicing/jointing, commissioning, AT, ABD etc.	Km	2,270		
5	Pulling/laying / blowing of optical Fibre Cable (144 core fiber) inside laid PLB duct and at crossings(with protection provided) ,splicing/jointing, commissioning, AT, ABD etc.	Km	868		
6	Pulling/laying / blowing of optical Fibre Cable (288 core fiber) inside laid PLB duct and at crossings(with protection provided) , splicing/jointing, commissioning, AT, ABD etc.	Km	375		



Annexure 1 (BOQ and Price) -LOA Amendment - 3

S. No	Item Description	Unit	AO-3 Qty	Unit Cost (INR)	AO-3 cost(INR)
B.10	Services- Active Infrastructure				
1	Installation, integration, testing and commissioning of Electronics at Zonal Level	No	3		
2	Installation, integration, testing and commissioning of Electronics at Mandal Level	No	182		
3	Installation, integration, testing and commissioning of Electronics at GP Level	No	3,775		
4	One Time Site Preparation cost for all Civil & Electrical Works including proper earthing at Zone	No	3		
5	One Time Site Preparation cost for all Civil & Electrical Works including proper earthing at Mandal	No	182		
6	One Time Site Preparation cost for all Civil & Electrical Works including proper earthing at GP	No	3,775		
B.11	Services for Aerial OFC Laying				
1	Erection of poles including excavation, fixing of MUFF, concreting, Backfilling, Reinstatement, Compaction of earth, Stay arrangements, coil earthing etc. and sign writing on poles with paint	No	3,882		
2	Sliding / Laying of ADSS 48 core OFC including fixing of installation accessories and fixtures etc., splicing, jointing, commissioning and AT etc.	Km	537		
3	Sliding / Laying of 2 core Drop cable (Oval Type) from splitter to the Govt. Inst. including survey, fixing of installation accessories and fixtures etc., splicing, jointing, commissioning and AT	Km	7,905		
B.12	Accessories at PoPs				
1	Fibre Patch cord (1M)	No	17,964		
2	Fibre Patch cord (3M)	No	32,348		
3	Fibre Patch cord (5M)	No	4,510		
4	Fibre Patch cord (10M)	No	1,100		
5	LGX 1:2 splitter (for aerial use)	Nos	263		
6	LGX 1:4 splitter (for aerial use)	Nos	651		
7	LGX 1:8 splitter (for aerial use)	Nos	5,449		
8	LGX 1:16 splitter (for aerial use)	Nos	161		
9	Splitter box type-1 (upto LGX 1:8 splitter)	Nos			
10	Splitter box type-1 (upto LGX 1:16 splitter)	Nos	143		
	Total of B (SHQ to GP) (CAPEX)				
	O&M except NOC & DWDM				
1	O&M per Route km of Core Ring i.e Zone to Zone Connectivity Year 1	Km	1,051.2		
	O&M per Route km of Core Ring i.e Zone to Zone Connectivity Year 2	Km	1,051.2		
	O&M per Route km of Core Ring i.e Zone to Zone Connectivity Year 3	Km	1,051.2		
	O&M per Route km of Core Ring i.e Zone to Zone Connectivity Year 4	Km	1,051.2		
	O&M per Route km of Core Ring i.e Zone to Zone Connectivity Year 5	Km	1,051.2		
	O&M per Route km of Core Ring i.e Zone to Zone Connectivity Year 6	Km	1,051.2		
	O&M per Route km of Core Ring i.e Zone to Zone Connectivity Year 7	Km	1,051.2		
2	O&M per Route km of Mandal to Mandal Connectivity Year 1	Km	2,392.58		
	O&M per Route km of Mandal to Mandal Connectivity Year 2	Km	2,392.58		
	O&M per Route km of Mandal to Mandal Connectivity Year 3	Km	2,392.58		
	O&M per Route km of Mandal to Mandal Connectivity Year 4	Km	2,392.58		
	O&M per Route km of Mandal to Mandal Connectivity Year 5	Km	2,392.58		
	O&M per Route km of Mandal to Mandal Connectivity Year 6	Km	2,392.58		
	O&M per Route km of Mandal to Mandal Connectivity Year 7	Km	2,392.58		
3	O&M per Route km of GP to GP Connectivity Year 1	Km	7,600.4		
	O&M per Route km of GP to GP Connectivity Year 2	Km	7,600.4		
	O&M per Route km of GP to GP Connectivity Year 3	Km	7,600.4		
	O&M per Route km of GP to GP Connectivity Year 4	Km	7,600.4		
	O&M per Route km of GP to GP Connectivity Year 5	Km	7,600.4		
	O&M per Route km of GP to GP Connectivity Year 6	Km	7,600.4		
	O&M per Route km of GP to GP Connectivity Year 7	Km	7,600.4		
4	O&M cost of each GP PoP including all active, passive & auxiliary infra Year 1	No.	3,775		
	O&M cost of each GP PoP including all active, passive & auxiliary infra Year 2	No.	3,775		
	O&M cost of each GP PoP including all active, passive & auxiliary infra Year 3	No.	3,775		



Annexure 1 (BOQ and Price) -LOA Amendment - 3

S. No	Item Description	Unit	AO-3 Qty	Unit Cost (INR)	AO-3 cost(INR)
	O&M cost of each GP PoP including all active, passive & auxiliary infra Year 4	No.	3,775		
	O&M cost of each GP PoP including all active, passive & auxiliary infra Year 5	No.	3,775		
	O&M cost of each GP PoP including all active, passive & auxiliary infra Year 6	No.	3,775		
	O&M cost of each GP PoP including all active, passive & auxiliary infra Year 7	No.	3,775		
5	O&M cost of each Mandal PoP including all active, passive & auxiliary infra Year 1	No.	182		
	O&M cost of each Mandal PoP including all active, passive & auxiliary infra Year 2	No.	182		
	O&M cost of each Mandal PoP including all active, passive & auxiliary infra Year 3	No.	182		
	O&M cost of each Mandal PoP including all active, passive & auxiliary infra Year 4	No.	182		
	O&M cost of each Mandal PoP including all active, passive & auxiliary infra Year 5	No.	182		
	O&M cost of each Mandal PoP including all active, passive & auxiliary infra Year 6	No.	182		
	O&M cost of each Mandal PoP including all active, passive & auxiliary infra Year 7	No.	182		
6	O&M cost of each Zonal PoP including all active, passive & auxiliary infra Year 1	No.	3		
	O&M cost of each Zonal PoP including all active, passive & auxiliary infra Year 2	No.	3		
	O&M cost of each Zonal PoP including all active, passive & auxiliary infra Year 3	No.	3		
	O&M cost of each Zonal PoP including all active, passive & auxiliary infra Year 4	No.	3		
	O&M cost of each Zonal PoP including all active, passive & auxiliary infra Year 5	No.	3		
	O&M cost of each Zonal PoP including all active, passive & auxiliary infra Year 6	No.	3		
	O&M cost of each Zonal PoP including all active, passive & auxiliary infra Year 7	No.	3		
7	O&M cost of each Govt. Institution including all active, passive & auxiliary infra Year 1	Km	7,905		
	O&M cost of each Govt. Institution including all active, passive & auxiliary infra Year 2	Km	7,905		
	O&M cost of each Govt. Institution including all active, passive & auxiliary infra Year 3	Km	7,905		
	O&M cost of each Govt. Institution including all active, passive & auxiliary infra Year 4	Km	7,905		
	O&M cost of each Govt. Institution including all active, passive & auxiliary infra Year 5	Km	7,905		
	O&M cost of each Govt. Institution including all active, passive & auxiliary infra Year 6	Km	7,905		
	O&M cost of each Govt. Institution including all active, passive & auxiliary infra Year 7	Km	7,905		
	Total of O&M for SHQ to GP (Except NOC + DWDM)				2,13,59,59,252.35
	Total of Capex & Opex for Phase 1 - Package C				9,13,80,01,701.01



CFA Cell
BSNL Corporate Office, 1st Floor
Bharat Sanchar Bhavan, Janpath, New
Delhi-110 001
Ph. 011-23714242



भारत संचार निगम लिमिटेड
(भारत सरकार का उपक्रम)
BHARAT SANCHAR NIGAM LIMITED
(A Govt. of India Enterprise)

No. BSNLCO-CFAO/17(15)/1/2020-CFA

Dated: 01.09.2021

To
M/s QuadGen Wireless Solutions P. Ltd.

Subject: Completion/Experience Certificates For Projects done For BSNL in Kerala and as HSSP for Bulk Users.

Ref: Your Letter No: NIL dated: 13.07.2021

1. With reference to above mentioned letter on noted subject please find attached the required certificate.
2. This has the approval of Competent Authority.

Encls: As Above


(P. Padmanabha Rao)
DGM (CFA)



CFA Cell
BSNL Corporate Office, 1st Floor
Bharat Sanchar Bhavan, Janpath, New
Delhi-110 001
Ph. 011-23714242



भारत संचार निगम लिमिटेड
(भारत सरकार का उपक्रम)
BHARAT SANCHAR NIGAM LIMITED
(A Govt. of India Enterprise)

No. BSNLCO-CFAO/17(15)/1/2020-CFA

Dated: 01.09.2021

TO WHOM SO EVER IT MAY CONCERN

This is to certify that M/s QuadGen Wireless Solutions Pvt. Ltd (PAN- AAACQ2351D, GSTIN is 29AAACQ2351D1ZQ) having registered office #607 World Trade Centre, Brigade Gateway, Malleshwaram West, Bangalore - 560055 is BSNL's Hotspot Services Provider (HSSP) Partner for Setting up, own and operate Wi-Fi Hotspots at public places through BSNL POP on revenue share basis, for Work order No: CE/WO/006/2014-15 dated 27.11.2014 for South Zone and Work order No: CE/WO/007/2014-15 dated 27.11.2014 for West Zone issued against EOI reference number CA/NWP-BB/Wi-Fi Hotspot/EOI-452/2013 dated 19.11.2013.

As on date, M/s QuadGen is successfully operating more than 7600+ Carrier Grade Wi-Fi Access Points, IPMPLS L2 Switch, Rack, UPS & Batteries connected to Centralized Wi-Fi Core having Core L3 Switch, Wireless Access Controller (WAC) of multiple OEMs, OSS/BSS, AAA, Firewall, Rack in BSNL NOC serving subscriber base more than 1 lakh Users spread across different locations.

These 7600+ also include the 4090 Wi-Fi Access Points working across Kerala circle under various work orders from Kerala State Information Technology Mission (KSITM).


(P. Padmanabha Rao)
DGM (CFA)



Enterprise Business Cell
O/o The Chief General Manager Telecom
Kerala Circle,
Trivandrum-33



भारत संचार निगम लिमिटेड
(भारत सरकार का उपक्रम)
BHARAT SANCHAR NIGAM LIMITED
(A Govt. of India Enterprise)

EB/10-19/2017-18/Part II/2

dated at Trivandrum the 18-12-2017
19

To

Sri. Satish Benarji,
Quadgen Wireless Solutions Pvt. Ltd.,
Bangalore.

Sir,

**Sub: Work Order for implementing 1000 WiFi hotspots across Kerala in Phase I
Of Public WiFi project of State Government.**

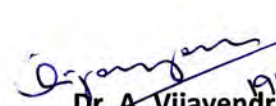
Ref: KSITM/Pwifi/1/2016/1521 dated 13-12-2017.

Kindly refer to the Work Order issued by IT Mission to BSNL, Kerala circle, for implementing 1000 WiFi hotspots across Kerala state as Phase I of 2000 WiFi hotspots project (copy enclosed).

In this regard, as the HSSP for the implementation of the project, it is requested to take necessary action as conveyed in RFP No. KSITM/Pwifi/1/2016 dated 31-7-2017 and the Work Order referred above to complete the Phase I within the timeline. Further, as part of project planning, it is requested that action plan with regard to supply, installation and commissioning of hotspot infrastructure at 1000 locations, with timelines, may be sent to this office immediately for the appraisal of CGMT, Kerala circle. A comprehensive survey report of the above mentioned locations, clearly indicating the requirement of APs and any additional item at each location, as mentioned in the RFP, along with other site related details, may also be submitted at the earliest.

It is also intimated that on the basis of the terms and conditions laid down in the RFP and Work order mentioned above, an agreement needs to be signed between BSNL, Kerala circle and M/s Quadgen with regard to Phase I implementation of the project. The date of signing of the agreement will be communicated in due course.

Yours Sincerely,


19/12/2017
Dr. A. Vijayendra,

Deputy General Manager (EB)
Office of Chief General Manager,
BSNL, Kerala Telecom Circle,
Thiruvananthapuram-33.





KSITM/Pwifi/1/2016/1521

13.12.2017

To,

The Chief General Manager,
Bharat Sanchar Nigam Limited,
Thiruvananthapuram

Sir,

Sub : Work Order for setting up 1000 Wifi Hotspots across Kerala for Phase-I

Ref : (1) e-Tender Ref No. KSITM/Pwifi/1/2016 dated 31.07.2017 (Tender Id:
2017_KSITM_135479_1, 2017_KSITM_135574_1, 2017_KSITM_135598_1)
(2) Letter of Intent (LOI) dated 20.09.2017 and its acceptance dated 28.10.2017

With reference to the tender and LoI issued, M/s Bharat Sanchar Nigam Limited is hereby informed to supply, install, commission the equipments and bandwidth for the setting up and maintenance of 1000 Wi-Fi Hotspots (Phase I) for a period of three years as per the mentioned terms and conditions.

The cost breakup for the components is listed below:

Sl.no	Period	Item Description	No. of Locations	Unit Price (Excl. Taxes)	Amount (Excluding Taxes)
1	Opex Charges for First year	Infrastructure Component	1000		
		Operations and maintenance	1000		
		Bandwidth charges for 10 Mbps (1:1)	1000		
2	Opex Charges for second year	Infrastructure Component	1000		
		Operations and maintenance	1000		
		Bandwidth charges for 10 Mbps (1:1)	1000		

3	Opex Charges for third year	Infrastructure Component	1000	
		Operations and maintenance	1000	
		Bandwidth charges for 10 Mbps (1:1)	1000	
Total Amount for 3 years				20,24,00,000.00

Optional Items

Sl.no	Item description	Unit Rate
1	Additional bandwidth charges for 10 Mbps (1:1)	
2	Additional bandwidth charges for 20 Mbps (1:1)	
3	Additional Access Point (Outdoor) at Hotspot with required passive cabling (with 80 mtr Armoured CAT 6 / OFC with media converter)	
4	Additional Access Point (Indoor) at Hotspot with required passive cabling (with 80 mtr Armoured CAT 6 / OFC with media converter)	
5	Additional 4 mtr GI pole (2.5 inch diameter, 2.5 mm or higher thickness - ISI) with required mounting box and accessories including installation charges with cement concrete base for outdoor installation.	
6	Additional Edge Switch	

Terms and Conditions

1. The scope of work, functional requirements, payment terms, timelines, deliverables, SLA, terms and conditions shall be as per the RFP and its amendments issued by KSITM.
2. The payment will be made in quarterly basis for each year and the payment will be released after the end of each completed quarter on submission of invoices and deliverables as per RFP.
3. The amount mentioned in this work order is excluding of Taxes. Taxes shall be paid by KSITM at actual rates applicable at the time of billing.
4. The site survey report shall be submitted by BSNL which will be finalized by KSITM.



5. The period of contract may be extendable to another 2 years based on mutual agreement at the rates arrived through the tender for AMC and Operations and Maintenance.
6. The make and model of the devices supplied shall be as follows:
Indoor Access Points - Cisco 1852
Outdoor Access Points - Cisco 1542
WLC Controller - Cisco 8540
Edge Switch - Dlink DGS-1210-10MP
7. The optional items will be ordered by KSITM on requirement basis during the period of contract. BSNL shall provide the required items as per the rates arrived through tender. The quantities of items will be as per KSITM requirement and the billing will be based on actual.
8. Any addition / change in the Hotspot locations, shifting shall be informed to KSITM and take prior approval.
9. BSNL shall execute an agreement with KSITM.

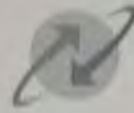
Samba Rao 7.

Director

SEERAM SAMBASIVA RAO IAS
DIRECTOR
Kerala State IT Mission
Department of Electronics & Information Technology



Enterprise Business Cell
O/o The Chief General Manager Telecom
Kerala Circle,
Trivandrum-33



भारत संचार निगम लिमिटेड
(भारत सरकार का उपक्रम)
BHARAT SANCHAR NIGAM LIMITED
(A Govt. of India Enterprise)

EB/10-19/2017-18/Part III/31

dated at Trivandrum the 18-9-2018

To

Sri. Satish Benarji,
Quadgen Wireless Solutions Pvt. Ltd.,
Bangalore.

Sir,

**Sub: Work Order for implementing 1000 WiFi hotspots across Kerala in Phase II
Of Public WiFi project of State Government.**

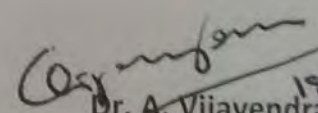
Ref: KSITM/Pwifi/2016/2563 dated 30-6-2018.

Kindly refer to the Work Order issued by IT Mission to BSNL, Kerala circle, for implementing 1000 WiFi hotspots across Kerala state as Phase II of 2000 WiFi hotspots project (copy enclosed).

In this regard, as the HSSP for the implementation of the project, it is requested to take necessary action as conveyed in RFP No. KSITM/Pwifi/1/2016 dated 31-7-2017 and the Work Order referred above to complete the Phase II within the timeline. Further, as part of project planning, it is requested that action plan with regard to supply, installation and commissioning of hotspot infrastructure at 1000 locations, with timelines, may be sent to this office immediately for the appraisal of CGMT, Kerala circle. A comprehensive survey report of the above mentioned locations, clearly indicating the requirement of APs and any additional item at each location, as mentioned in the RFP, along with other site related details, may also be submitted at the earliest.

It is also intimated that on the basis of the terms and conditions laid down in the RFP and Work order mentioned above, a PBG worth Rs. 1.012 Cr. shall be submitted to BSNL, Kerala circle and an agreement needs to be signed between BSNL, Kerala circle and M/s Quadgen. The date of signing of the agreement will be communicated in due course.

Yours Sincerely,


Dr. A. Vijayendra,
Deputy General Manager (EB)
Office of Chief General Manager,
BSNL, Kerala Telecom Circle,
Thiruvananthapuram-33.





Kerala State Information Technology Mission
Department of Electronics & Information Technology
Government of Kerala

ICT Campus, Vellayambalam Jn.,
Thiruvananthapuram 695 033, Kerala, India
Tel: +91 -471-2726881, 2314307, Telefax: +91-471-2314284
Email: admin.ksitm@kerala.gov.in

KSITM/Pwifi/2016/ **2563**
30.06.2018

To,

The Chief General Manager,
Bharat Sanchar Nigam Limited,
Thiruvananthapuram

Sir,

Sub : Work Order for setting up 1000 Wifi Hotspots across Kerala for Phase-II

Ref : (1) e-Tender Ref No. KSITM/Pwifi/1/2016 dated 31.07.2017 (Tender Id:
2017_KSITM_135479_1, 2017_KSITM_135574_1, 2017_KSITM_135598_1)
(2) Work Order for Phase -I dated 13.12.2017

With reference to the tender referred above, M/s Bharat Sanchar Nigam Limited is hereby informed to supply, install and commission the equipments and bandwidth for the setting up and maintenance of 1000 Wi-Fi Hotspots (Phase II) for a period of three years as per the below terms and conditions.

The cost breakup for the components is listed below:

Sl.no	Period	Item Description	No. of Locations	Unit Price (Excl. Taxes)	Amount (Excluding Taxes)
1	Opex Charges for First year	Infrastructure Component	1000		
		Operations and maintenance	1000		
		Bandwidth charges for 10 Mbps (1:1)	1000		
2	Opex Charges for second year	Infrastructure Component	1000		
		Operations and maintenance	1000		
		Bandwidth charges for 10 Mbps (1:1)	1000		
3	Opex Charges	Infrastructure Component	1000		

	for third year	Operations and maintenance	1000		
		Bandwidth charges for 10 Mbps (1:1)	1000		
Total Amount for 3 years					20,24,00,000.00

Optional Items

Sl.no	Item description	Unit Rate
1	Additional bandwidth charges for 10 Mbps (1:1)	
2	Additional bandwidth charges for 20 Mbps (1:1)	
3	Additional Access Point (Outdoor) at Hotspot with required passive cabling (with 80 mtr Armoured CAT 6 / OFC with media converter)	
4	Additional Access Point (Indoor) at Hotspot with required passive cabling (with 80 mtr Armoured CAT 6 / OFC with media converter)	
5	Additional 4 mtr GI pole (2.5 inch diameter, 2.5 mm or higher thickness - ISI) with required mounting box and accessories including installation charges with cement concrete base for outdoor installation.	
6	Additional Edge Switch	

Terms and Conditions

1. The scope of work, functional requirements, payment terms, timelines, deliverables, SLA, terms and conditions shall be as per the RFP and its amendments issued by KSITM.
2. The payment will be made in quarterly basis for each year and the payment will be released after the end of each completed quarter on submission of invoices and deliverables as per RFP.
3. The amount mentioned in this work order is excluding of Taxes. Taxes shall be paid by KSITM at actual rates applicable at the time of billing.
4. The site survey report shall be submitted to KSITM including the Coastal Villages on priority basis for the establishment of hotspots.
5. The period of contract may be extendable to another 2 years based on mutual agreement at the rates arrived through the tender for AMC and Operations and Maintenance.
6. The make and model of the devices supplied shall be as follows:
 - Indoor Access Points – Cisco 1852
 - Outdoor Access Points – Cisco 1542
 - WLC Controller - Cisco 8540



- Edge Switch - Dlink DGS-1210-10MP
7. The optional items mentioned will be ordered by KSITM on requirement basis during the period of contract. BSNL shall provide the required items as per the mentioned rates. The quantities of items will be as per KSITM requirement and the billing will be based on actual.
 8. Any addition / change in the Hotspot locations, shifting shall be informed to KSITM and take prior approval.
 9. BSNL shall execute an agreement with KSITM

Sarabathu J.

Director
SEETHA RAMBASIVA RAO IAS
DIRECTOR
 Kerala State IT Mission
 Department of Electronics & Information Technology

